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Introduction

I'm sure there are plenty of FRG-8800's out there with one or more dead dial lamps. Now while this doesn't stop the set from working it sure spoils the look of it. My FRG-8800 had all lamps out. I didn't really want to send it away for repair so I thought I'd have a crack at it myself.

Before we go any further I warn you that this procedure requires electronics knowledge and manual skills including the use of a soldering iron. If you are not electronically inclined or if you're not confident with a soldering iron then stop here. If you continue on and follow these instructions and something goes wrong I can't help you and I won't be held responsible for any damage you might cause to your set.

The FRG-8800 uses four miniature grain of wheat lamps to illuminate the display. These lamps have a little green rubber sock placed over them to get some color into the display. I had a look in the local electronic stores here in Australia* for suitable replacements but the closest I could find was 4x10mm 6V globes with flying leads. Now whilst that was OK I thought I could do a bit better.

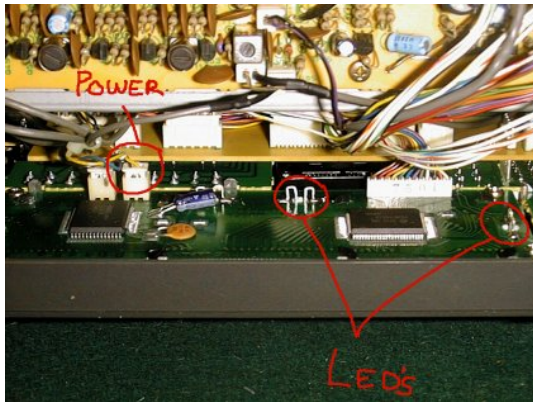
I only wanted to do this once and not have to think about changing lamps in the future so I thought I would use leds instead of incandescent lamps. Now whilst the green display looks OK and I'm sure there are many arguments regarding color selection to minimize eye fatigue etc. I don't expect to be staring at the display for hours on end. I decided to give the

set a cosmetic boost and settled on the currently very fashionable blue for the display. I went out and purchased four high brightness 3mm blue leds. These leds are rated at 1000 mcd and are quite stunning.



The display assembly is quite easy to access and with a little care it is

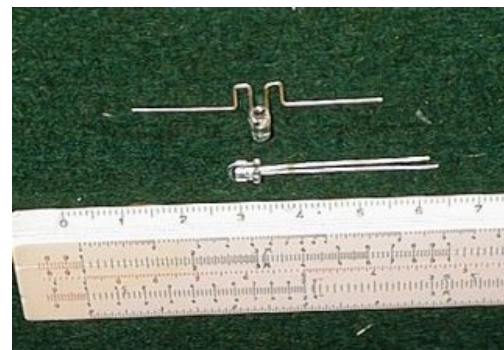
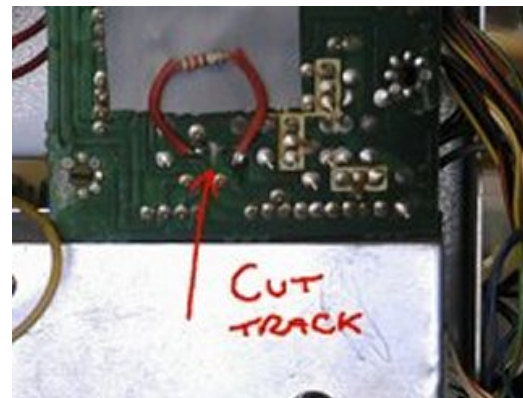
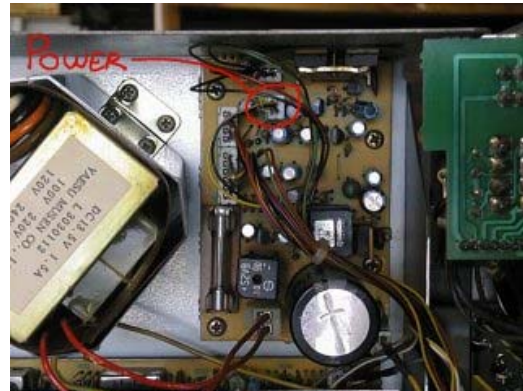
possible to remove the old lamps and install the new leds without taking the display assembly out of the radio. The display lamps are all in parallel and the open circuit supply voltage is around 7.3 volts. This is supplied via the black and yellow wires that plug in to the back of the display board. These wires run to the power supply pcb.



I don't have a service manual for the set so I had to trace the wiring out to see how it worked. A bit of snooping about with a multi-meter showed that the dimmer switch toggled a control voltage on the supply pcb which probably meant there was a transistor involved somewhere. Tracing the circuit a little it seemed that there was more than one transistor involved in the process and as it didn't look like the last transistor in the chain had a collector resistor I decided to put a 330 ohm resistor in series with the leds. This gave a total led current of 12mA which was just fine.

I cut the track leading from the collector of the transistor to the yellow lead and put the resistor in series with it. A bit of tape over the tracks and some sleeving for the

resistor legs makes sure there are no shorts.



I took a bit of time bending the leads for the leds to make sure that there was plenty of length to absorb some of the heat during soldering. I've never had much success with soldering very close to the body of a 3mm led. I've found the closest I can get is about 10 - 12mm from the body ever with a 600 degree tip. Anyway this is how I bent the leads

and as it turned out it made it easy to hold whilst soldering and the extra lead length makes it easy to push the led centrally into the hole. Make sure the led goes all the way into the hole.



The finished product looks magnificent and the blue is particularly striking. It took about an hour to complete but the results are worth it. The picture doesn't really do it justice. The dark patches you see

in this image are probably due to the camera. Whilst the dark areas are visible by eye they are not as prominent as in the image.

* For the information of anyone in Australia that might want to try this Dick Smith no longer stock the grain of wheat globes and all they had at my local Powerhouse were 12V or 1.5V globes with flying leads. Jaycar had 4x10mm 6V globes with flying leads(SL2673). Jaycar also had the 3mm blue leds(ZD0130).

Disclaimer

This project is offered as is with no guarantees and with the understanding that it may contain errors.

I assume that the reader will understand the technical details and are capable of building and installing this modification without issue.

Should you break your radio, remember who took the covers off and started this activity. I am not responsible for you breaking your radio. Do not start this project unless you are sure you are capable of doing the work.